

# Reihaneh Malekian

Ph.D. Candidate in Statistics, Columbia University  
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## Education

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**Columbia University**, New York, NY

Expected May 2027

Ph.D. in Statistics

Advisor: Sumit Mukherjee

**Sharif University of Technology**, Tehran, Iran

M.Sc. in Pure Mathematics

**Sharif University of Technology**, Tehran, Iran

B.Sc. in Mathematics and Applications

## Research Interests

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Probability and high-dimensional statistics; large deviations; random graphs and hypergraphs; Gibbs measures and interacting systems; empirical Bayes; statistical inference for dependent data.

## Publications and Preprints

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- **Reihaneh Malekian**, Hao Xing, and Arian Maleki. “Is Speckle Noise More Challenging to Mitigate Than Additive Noise?” *IEEE Transactions on Information Theory*, 71(8):6353–6372, 2025.
- **Reihaneh Malekian** and Aaditya Ramdas. “Matrix Concentration: Order versus Anti-order.” Preprint, arXiv:2408.05998.

## Manuscripts in Preparation

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- “LDP (Large Deviation Principles) for Tensor Forms,” with Sohom Bhattacharya, Nabarun Deb, and Sumit Mukherjee.
- “Fluctuations in Mean-Field Potts Models,” with Sumit Mukherjee.

## Selected Research Presentations

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- Invited talk, Probability Seminar, UC Berkeley, September 2026.
- Contributed talk, Joint Statistical Meetings, Boston, August 2026.
- Talk, ICERM Hot Topics Workshop, Brown University, August 2026.
- Poster, International Congress of Mathematicians, Philadelphia, July 2026.
- Poster, Peter Hall and Bob Shumway Memorial Conference on Dependent Data Analysis in the Modern Era, UC Davis, May 2026.
- Invited talk, Finger Lakes Probability Seminar, Syracuse University, April 2026.
- Seminar talk, Columbia University Student Seminar, February 2026.
- Talk, Progress in Discrete and Continuous Probability Conference, Union College, Schenectady, NY, July 2025.
- Poster, INFORMS Applied Probability Society Conference, Atlanta, July 2025.
- Poster, Bernoulli–IMS World Congress in Probability and Statistics, Bochum, Germany, August 2024.
- Talk, Minghui Yu Memorial Conference, Columbia University, April 2024, April 2025 and April 2026.

## Honors and Awards

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- Teaching Observation Fellow, Columbia University Center for Teaching and Learning, 2026–2027.
- Early Career Travel Award, American Statistical Association, 2026.
- Graduate Research Fellowship, Columbia University, 2022–2027.
- Ranked second in overall GPA in both the B.Sc. and M.Sc. cohorts, Sharif University of Technology.
- Recipient of undergraduate and graduate scholarships from Iran’s National Elites Foundation.
- Admitted to the graduate program without an entrance examination as an exceptionally talented B.Sc. student, Sharif University of Technology.
- Placed among the top 0.05% of participants in Iran’s nationwide university entrance examination.

## Selected Workshops and Summer Schools

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### ICERM Hot Topics Workshop

Synergies between Geometry, Probability, and Computation in High Dimensions, Brown University August 2026

### SLMath Summer Graduate School

Concentration Inequalities and Localization Techniques in High-Dimensional Probability and Geometry July 2023

## Teaching Experience

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**Teaching Assistant**, Columbia University 2022–Present

- Introduction to Statistics; Statistical Inference; Linear Regression Models; Introduction to Statistical Reasoning; Introduction to Probability and Statistics; Probability.

**Teaching Assistant**, Sharif University of Technology

- Probability and Applications; Probability Theory; Mathematical Writing; Algebra I; Analysis I.

## Research Mentoring

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**Bachelor’s Project Co-supervisor**, EPFL 2021

Co-supervised Clémence Marcaillou’s bachelor’s project in combinatorial discrepancy theory through weekly research meetings.

## Selected Expository Writing

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- **Reihaneh Malekian**. “Good Numbers.” *Pargar*, Year 2, no. 4, pp. 17–23, Spring 2014. [In Persian]
- **Reihaneh Malekian** (translator and adapter). “The Turán Number and Probabilistic Combinatorics,” by Alan J. Aw. Persian translation and adaptation of the article published in *The American Mathematical Monthly*, 119(6):510–513, 2012. *Pargar*, Year 2, no. 8, pp. 9–13, Spring 2015. [In Persian]

## Outreach and Academic Service

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**Organizing Team Member and Program Moderator, Maryam Mirzakhani Celebration** May 2019

Sharif University of Technology, Tehran, Iran

**Student Organizer**, Undergraduate Mathematics Summer School, IASBS, Zanjan, Iran 2019

**Student Organizer**, Mathematics Summer School for High School Students, Borujen, Iran 2015

## Skills

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**Programming:** Python, R.

**Languages:** Persian (Farsi; native), English (proficient), French (intermediate, B1).